Semantic Grounding and Safe Execution for PostGIS Natural-Language-to-SQL

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Abstract

Natural language interfaces to geospatial databases must handle challenges that mainstream text-to-SQL benchmarks rarely exercise: geometry-aware schema grounding, OGC/PostGIS topological and metric predicates, coordinate reference systems, `::geography` casting, KNN operators, and safety-critical refusal of destructive or unbounded queries. We present **NL2Semantic2SQL**, a semantic-grounding and safe-execution framework for executable PostGIS NL2GeoSQL. It couples a semantic layer (aliases, geometry types, SRIDs, spatial operator rules) with an intent-conditioned router, schema-aware prompting, few-shot retrieval, a SQL postprocessor that enforces read-only safety and injects `LIMIT` on large-table preview queries, and an execution-time self-correction loop. The same bilingual architecture serves Chinese GIS and English warehouse queries from a single pipeline; we evaluate both sides so that the current state of each is measurable. We evaluate on a 125-question Chongqing PostGIS benchmark (85 Spatial + 40 Robustness across six safety categories) and on a warehouse evaluation track on BIRD mini_dev, reported as a 108-question design set and a disjoint 150-question held-out set. On GIS Spatial we pool three independent Full-mode runs (each 85 questions) and report mean $\pm$ SD EX$= 0.663 \pm 0.048$ (range $[0.612, 0.706]$) vs. baseline 0.529; the majority-vote Full result is 0.659, with paired McNemar $p = 0.052$ (marginal). Two of three individual samples reach significance ($p = 0.003$, $p = 0.029$); one does not. The 40-question Robustness suite is the strongest primary claim: Full reaches $39/40 = 0.975$ vs. baseline 0.450 (paired McNemar $b = 0$, $c = 21$, $p < 10^{-4}$), with OOM Prevention bounded-answer compliance improving from $1/8$ to $7/8$ after the `EXPLAIN`-based pre-check and a revised bounded-output agent prompt. On BIRD, the pipeline improves the design set by $+0.093$ ($p = 0.0213$) but only $+0.033$ on the held-out set ($p = 0.3833$, 95% CI $[-0.03, 0.09]$); this current-state snapshot motivates a planned second round of warehouse-side grounding. A 50-question Chinese BIRD probe is additionally reported alongside a bilingual human re-audit (18 OK, 32 FIX, 0 DROP; post-review EX 0.540 vs. pre-review 0.560, paired $p = 1.00$) that rules out translation artefacts as the cause of the English→Chinese gap.

Keywords: NL2GeoSQL; PostGIS; natural language interface; geospatial database; semantic layer; spatial predicates; coordinate reference systems; safe execution; large language model.

1 Introduction

Natural language access to spatial databases is a long-standing GIScience problem [1, 2, 3]. Recent large language models have pushed general text-to-SQL forward on benchmarks such as Spider [15] and BIRD [20], yet these benchmarks deliberately abstract away the execution machinery that defines geospatial querying: the OGC Simple Feature Access predicates [4], coordinate reference system transformations, the `geometry`/`geography` type distinction, spatial indexes, KNN operators, and the safety constraints that any production GIS must enforce against destructive or unbounded queries. Natural language to executable PostGIS [6] (NL2GeoSQL) is therefore not an instance of text-to-SQL with a GIS dataset but a distinct problem shaped by spatial-database semantics.

The grounding stage is where the gap is most visible. General text-to-SQL schema linking aligns user expressions with column names; PostGIS grounding must additionally recognise geometry-bearing attributes, infer whether a question requires a topological predicate (`ST_Intersects`, `ST_Touches`, `ST_Contains`), a metric operator (`ST_Distance`, `ST_DWithin`, `ST_Area`), a buffer or KNN operator (`<->`), and decide when a geography cast or `ST_Transform` is required so that area and distance are returned in metres rather than degrees. It must also refuse destructive intent (`DELETE`, `UPDATE`, `DROP`) and enforce bounded output on large-table preview queries. These are GIScience computations, not prompt-engineering problems.

We present **NL2Semantic2SQL**, a semantic-grounding and safe-execution framework for PostGIS-based NL2GeoSQL. A semantic layer provides aliases, geometry metadata, spatial operator rules, and safety constraints; a bilingual intent classifier routes Chinese GIS and English warehouse queries into the appropriate grounding path; a SQL postprocessor enforces read-only safety, repairs identifier quoting, and injects `LIMIT` on large-table preview queries; and an execution-time self-correction loop recovers from common PostGIS execution errors such as missing geography casts and mixed-case identifier quoting. The same framework serves both GIS and warehouse users, and we evaluate both sides so that the current state of each is measurable.

For primary evaluation, we construct a 125-question Chongqing PostGIS benchmark with executable gold SQL (85 Spatial covering OGC-Simple-Feature predicates, metric and KNN operators, CRS transforms, and aggregation across three difficulty levels; plus 40 Robustness spanning six safety and refusal categories). For the warehouse evaluation track, we use BIRD mini_dev with a deliberate design/held-out split: a 108-question *design set* used to derive grounding rules by error attribution, and a disjoint 150-question *held-out set* for independent significance. We additionally run a 50-question cross-lingual probe on LLM-translated Chinese BIRD questions together with a bilingual human re-audit and paired re-run on the reviewed subset, so that translation-pipeline noise and genuine cross-lingual capability can be distinguished.

Contributions. The principal contribution is PostGIS-side grounding; the BIRD evaluation is a second track that reports the current state of the warehouse path of the same framework.

1. **A semantic-grounding and safe-execution framework for NL2GeoSQL** that declares geometry types, SRIDs, spatial operator rules, and refusal constraints in a semantic layer and threads them through intent-conditioned routing, schema-aware prompting, SQL postprocessing, and execution-time self-correction.

2. **A 125-question Chongqing PostGIS benchmark** with executable gold SQL verified against a live PostGIS database, explicitly separating 85 Spatial questions (OGC predicates, metric and KNN operators, CRS transforms, aggregation) from 40 Robustness questions (six safety categories). On GIS Spatial, three independent Full-mode runs yield pooled mean $\pm$ SD EX$= 0.663 \pm 0.048$ (range $[0.612, 0.706]$) vs. baseline 0.529; two of three individual samples are paired-significant (paired McNemar $p = 0.003$, $p = 0.029$) and majority-vote $p = 0.052$ (marginal). The 40-question Robustness suite is the strongest primary claim: Full reaches $39/40 = 0.975$ vs. baseline $18/40 = 0.450$ (paired McNemar $b = 0$, $c = 21$, $p < 10^{-4}$), with OOM Prevention bounded-answer compliance improving from $1/8$ in v3 to $7/8$ in v4.

3. **A warehouse evaluation track on BIRD mini_dev** separating an 108-question design set (rule derivation) from a 150-question held-out set (independent significance). Grounding rules learned on the design set improve the design-set EX by $+0.093$ ($p = 0.0213$) but improve held-out EX by only $+0.033$ (95% CI $[-0.03, 0.09]$, $p = 0.3833$), a current-state snapshot that we use to scope the next round of warehouse-side grounding work rather than a closed generalisation claim.

The paper proceeds as follows. Section 2 reviews GIScience foundations and the recent IJGIS-published NL-to-spatial-database literature. Section 3 formalises the problem and details the architecture. Section 4 presents the PostGIS evaluation and the BIRD warehouse evaluation track. Section 5 discusses interpretation. Section 6 concludes.

Table 1: Direct comparison to recent NL-to-spatial-database systems. ✓: supported / reported; partial: some support; −: not reported.

System	Executable PostGIS	Execution-based eval	CRS / geog. / KNN	Safety / refusal	Released eval set	Paired signif.
NALSpatial [14]	✓	✓	partial	−	✓	−
Monkuu [13]	✓	✓	partial	−	−	−
GeoSQL-Eval [10]	−	✓	✓	−	✓	−
GeoCogent [11]	−	−	−	−	−	−
GeoAgent [12]	−	partial	partial	−	−	−
Ours	✓	✓	✓	✓	✓	✓

2 Related Work

2.1 GIScience foundations for NL-to-spatial-database

Natural language access to spatial databases sits within the GIScience programme articulated by Goodchild [1]: formalising geographic concepts so they can be computed rigorously. The relational spatial semantics we ground against have roots in the 4- and 9-intersection formalisms [2] and the simplified end-user topological relations of Clementini et al. [3] — equals, disjoint, touches, contains, overlaps, within, crosses, intersects — which the OGC Simple Feature Access standard codifies for relational spatial SQL [4] and GeoSPARQL [5] extends to RDF. PostGIS [6] operationalises these in PostgreSQL and defines the `geometry`/`geography` type distinction, the `::geography` cast, `ST_DWithin`, `ST_Transform`, and the `<->` KNN operator any PostGIS-generating system must respect. Our grounding rules encode these formalisms as prompt-side evidence.

2.2 Natural language interfaces to spatial databases

Early geospatial question answering focused on small fact databases: Zelle and Mooney [7] used inductive logic programming for GeoQuery, and Punjani et al. [8] extended this to template-based question answering over linked geospatial data, highlighting spatial-predicate coverage and coordinate-system awareness. Recent LLM-era work falls into two groups. The first targets end-to-end NL-to-executable-spatial-SQL: NALSpatial [14] translates user queries into spatial SQL over PostGIS-style backends; Monkuu [13] is an IJGIS-published LLM-powered NL interface with dynamic schema mapping; GeoSQL-Eval [10] contributes the first large-scale LLM evaluation on PostGIS NL2GeoSQL (14,178 tasks across 340 PostGIS functions at function level). The second targets higher-level GeoAI agents: GeoCogent [11] is an LLM-based agent for geospatial code generation; GeoAgent [12] is a hierarchical multi-agent architecture. Surveys map the broader opportunities [9]. Our contribution lies at the narrower interface between spatial-database semantics and executable SQL: we study end-to-end execution-graded NL-to-PostGIS-SQL with a semantic-layer grounding that explicitly encodes OGC predicates, SRIDs, geography casts, and KNN operators, together with a postprocessor enforcing read-only and bounded-output semantics. Compared with NALSpatial and Monkuu we contribute a 125-question execution benchmark with an explicit Robustness suite, paired McNemar significance testing, and a design/held-out split on a warehouse benchmark as a second track of the same bilingual framework.

The comparison in Table 1 summarises capability and evaluation practice across recent NL-to-spatial-database systems.

2.3 Text-to-SQL with LLMs and semantic layers

LLM text-to-SQL has progressed through Spider [15, 16], BIRD [20], Spider 2.0 [21], and BEAVER [22]. DIN-SQL [17] introduced decomposed in-context prompting with schema linking, query classification, generation, and self-correction; DAIL-SQL [18] introduced skeleton-similarity few-shot selection; C3 [19] explored zero-shot ChatGPT text-to-SQL with calibrated bias correction; CSpider [23] is an early multilingual benchmark. Semantic-layer thinking — entities, measures, dimensions, metric graphs — is operationalised in MetricFlow / dbt's semantic layer [24] for BI; we adapt it by combining it with GIS annotations (geometry types, SRIDs, OGC predicates). Here BIRD serves as a warehouse track of the same bilingual framework, reported as a design/held-out pair so the current warehouse path is measurable and the next engineering round scopable.

3 Method

3.1 Formal scope: semantic layer as metadata graph $G = (V, E)$

We formalise the semantic layer as a directed multigraph $G = (V, E)$ that encodes the executable subset of OGC Simple Feature Access [4] required for PostGIS grounding. The abstraction follows the metadata-graph proposal in prior reviewer feedback.

Vertices. $V = E_{geo} \cup D \cup M \cup I \cup P \cup C$:

- E_{geo} — geographic entities; each vertex is a PostGIS table with attributes (geometry column, SRID, geometry type).
- D, M — dimension and measure vertices for the warehouse track.
- I — intent tags: `attribute_filter`, `aggregation`, `spatial_join`, `spatial_measurement`, `knn`, `preview_listing`.
- P — OGC SFA predicate vertices, partitioned into topological {`ST_Intersects`, `ST_Contains`, `ST_Within`, `ST_Touches`, `ST_Crosses`, `ST_Overlaps`, `ST_Equals`}, metric {`ST_Distance`, `ST_Length`, `ST_Area`, `ST_DWithin`}, and KNN {`<->`}.
- C — constraint vertices for SRID ranges, unit rules, and safety/refusal.

Edges. E comprises $E_{hasGeom}$ (table ↔ geometry column), E_{fk} (foreign keys from `information_schema.key_colum`
$E_{topo}, E_{metric}, E_{knn}$ (predicate ↔ entity), E_{route} (intent → predicate subset), E_{safety} (destructive-intent refusal), and E_{unit} (SRID range → required cast → yielded unit). Algorithm 1 materialises G at agent startup from the live PostGIS catalog; our reference implementation is released as a single Python module that the runtime agent consumes before each grounding call.

Executable subset, not a complete ontology. G is not a complete geospatial ontology and is not a GeoSPARQL [5] knowledge base. It is an *executable subset* of the OGC SFA predicate taxonomy, expressed as prompt-side evidence for an LLM-driven NL2GeoSQL agent. Full ontology alignment (OGC SFA, ISO 19107, GeoSPARQL) is left to future work; Table 2 documents the rule-level mapping we already enforce, showing that each semantic-layer rule corresponds to a specific OGC SFA clause rather than borrowing the standard only by name.

3.2 Problem formulation

We formulate cross-domain natural language to SQL as the task of translating a user question q into an executable SQL statement s over heterogeneous structured databases that may belong to either geospatial or conventional warehouse domains. Let S denote the executable schema representation extracted from the database catalog, and let E denote semantic-side evidence such as aliases, hierarchy expansions, geometry

Algorithm 1 BUILDSEMANTICGRAPH($schema, prefix$)

```
G ← empty multigraph
for (t, c, srid, τ) ∈ geometry_columns do
    if t starts with prefix and srid ≠ 0 then
        add GEO_ENTITY vertex (t, c, srid, τ)
    end if
end for
for (t_s, t_t) ∈ key_column_usage do
    add edge (t_s → t_t, FK)
end for
register topological / metric / KNN predicate vertices (OGC SFA 06-104r4)
register intent vertices; route intents → predicate classes
register unit-rule edges: metric → SRID range → required cast
return G
```

Table 2: Alignment of semantic-layer rules in $G = (V, E)$ to OGC SFA (06-104r4) / ISO 19107 clauses and PostGIS functions. Rule-level, not concept-level — each enforced rule traces to a specific clause.

Rule	Edge class	OGC 06-104r4 clause	ISO 19107 clause	PostGIS
`R_topo_intersects`	E_{topo}	§6.1.2.3 (Relate)	§6.2.2	`ST_Intersects`
`R_topo_contains`	E_{topo}	§6.1.2.3 (Relate)	§6.2.2	`ST_Contains`
`R_topo_within`	E_{topo}	§6.1.2.3 (Relate)	§6.2.2	`ST_Within`
`R_metric_length_geog`	E_{unit}	§6.1.2.3 (Length) + §8.2.3 (units)	§6.4.14 (GM_Curve.length)	`ST_Length(geography)`
`R_metric_area_geog`	E_{unit}	§6.1.2.3 (Area) + §8.2.3 (units)	§6.4.15 (GM_Surface.area)	`ST_Area(geography)`
`R_metric_dwithin_geog`	E_{unit}	§6.1.2.3 (Distance) + §8.2.3	§6.4.10 (distance)	`ST_DWithin(geography)`
`R_knn_nearest`	E_{knn}	§6.1.2.3 (Distance)	§6.4.10	`<->` KNN operator
`R_safety_refusal`	E_{safety}	not in OGC	—	postprocessor AST guard
`R_safety_bounded`	E_{safety}	not in OGC	—	`LIMIT` injection

metadata, value hints, or fact/dimension/entity annotations. The problem can then be written abstractly as

$$\hat{s} = f(q, S, E), \tag{1}$$

where f is implemented by an LLM-conditioned generation pipeline rather than a single end-to-end model. Unlike standard text-to-SQL settings, the target databases in our problem formulation are not assumed to share homogeneous operator semantics. In warehouse-style databases, correct SQL generation primarily depends on schema linking, join reasoning, aggregation, and temporal filtering. In geospatial databases, however, the model must additionally reason about geometry-bearing columns, spatial relations, coordinate systems, and domain-specific operators such as intersection, containment, buffering, and area or distance calculation. The purpose of our framework is therefore not only to generate valid SQL, but also to provide a semantic mediation layer that adapts the grounding process to the structural characteristics of each domain.

3.3 Three-stage pipeline

We design NL2Semantic2SQL as a three-stage pipeline (Figure 1). In the first stage, the system performs semantic resolution over a semantic layer that stores table-level and column-level annotations, aliases, domain hints, and task-relevant metadata. In the second stage, the resolved information is combined with database schema inspection and optional few-shot retrieval to construct a grounded context block for SQL

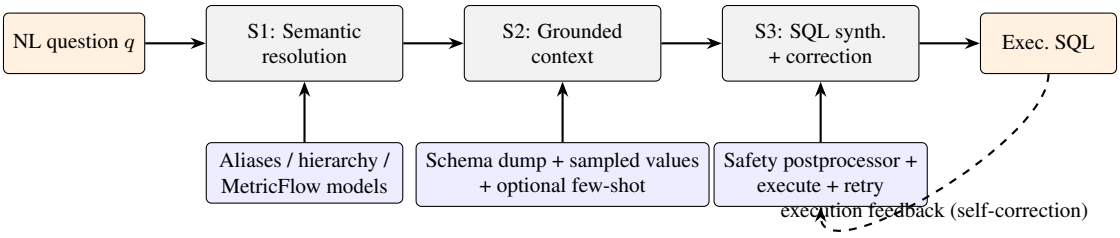

Figure 1: Architecture of NL2Semantic2SQL. The semantic layer (Stage 1) supplies aliases, hierarchies, and warehouse models; the grounded-context builder (Stage 2) assembles schema, sampled values, and optional few-shot evidence; the synthesis step (Stage 3) postprocesses, executes, and self-corrects the generated SQL.

generation. In the third stage, the generated SQL is postprocessed, executed under safety constraints, and, when necessary, revised through an execution-feedback self-correction loop. This decomposition separates semantic interpretation from SQL synthesis, allowing the framework to expose domain-specific evidence to the language model while keeping the final generation step constrained by executable schema information.

3.4 Geospatial-aware grounding and bilingual routing

The framework's geospatial path explicitly identifies geometry columns, geometry types, and spatial reference information, and injects operator-level rules into the grounding context. These rules distinguish topological from metric predicates and specify when area or distance calculations require coordinate transformation or geography casting, reducing the burden on the language model to infer execution constraints from raw schema text. When spatial signals are absent, the framework falls back to generic schema linking and aggregation-oriented prompt construction. Routing between these paths is driven by a bilingual intent classifier that recognises Chinese GIS keywords (spatial operator terms, Chinese administrative region names, Chinese measurement units) alongside English warehouse expressions ("how many", "average", "compare"). In real-world deployments GIS analysts often query in Chinese while warehouse analysts use English, and a single interface must handle both without language-specific model swapping; bilingual routing is therefore a first-class design choice rather than an afterthought.

3.5 Warehouse-side enhancements

For non-geospatial warehouse queries, the framework adds three domain-adaptive enhancements. *Cross-domain candidate ranking* reprioritises sources based on whether the parsed semantic context contains spatial operators or region filters; when such signals are absent, geometry-bearing tables are deprioritised and tables with matched column evidence are boosted. *Value-aware grounding* injects sample distinct values from low-cardinality text columns so that the model can use exact categorical values rather than hallucinated strings. *MetricFlow-style fact/dimension modelling* stores per-table roles, entity columns (primary or foreign join keys), and measure columns in a semantic-model registry; at grounding time, candidate tables are matched against this registry and explicit join-path hints are derived by aligning shared entities across fact/dimension pairs. These hints are injected only for non-spatial queries, leaving the GIS path unchanged.

3.6 Execution-time self-correction

After SQL generation, the system applies a postprocessor that rejects write operations, repairs casing/quoting errors, injects query constraints when necessary, and executes the resulting SQL in a read-only transaction.

Stage	Content
NL question (q)	*Chinese original, transliterated: tǒngjì suǒyǒu dānhángdào de zǒngchángdù (gōnglǐ). English gloss:* "Total length in kilometres of all one-way roads."
Intent classifier	`intent = spatial_measurement`, `metric = length`, `unit = km`, `language = zh`.
Semantic resolution (Stage 1)	Matches `cq_osm_roads_2021` as the roads table; resolves the Chinese alias for "one-way road" to the filter `oneway = 'yes'`; identifies `geometry` column (MultiLineString, SRID 4326, WGS 84 lat/lon).
Grounded context (Stage 2)	Schema dump for `cq_osm_roads_2021`; two injected rules: (i) *metric operators on lat/lon data require `::geography` cast so that `ST_Length` returns metres*; (ii) *unit conversion to km is a divide by* 1 000 *performed after `SUM`*. One spatial-measurement few-shot example with geography cast.
Initial SQL $s^{(0)}$	`SELECT ROUND(SUM(ST_Length(geometry)) / 1000.0, 2) AS length_km FROM cq_osm_roads_2021 WHERE oneway = 'yes';`
Postprocessor + execution	Executes, but returns a value in degree units because `::geography` was not applied; the postprocessor tags this as a "metric unit sanity" regression against the few-shot expectation and feeds the error message back to $g(\cdot)$.
Corrected SQL $s^{(1)}$	`SELECT ROUND((SUM(ST_Length(geometry::geography)) / 1000.0)::numeric, 2) AS length_km FROM cq_osm_roads_2021 WHERE oneway = 'yes';`
Gold SQL	`SELECT ROUND((SUM(ST_Length(geometry::geography)) / 1000.0)::numeric, 2) FROM cq_osm_roads_2021 WHERE oneway = 'yes';`
Outcome	Execution-based match (EX = 1); the postprocessor additionally adds the numeric cast required by PostgreSQL's `ROUND(double precision, integer)` resolution.

Figure 2: Worked example tracing a single Chongqing benchmark question through the full pipeline. The baseline LLM (not shown) omits the `::geography` cast and returns a value in degrees; our Stage 2 rule "metric operators on lat/lon data require `::geography`" is what closes the gap.

If execution fails, the framework enters a bounded self-correction loop. Let $s^{(0)}$ be the initial generated SQL and let $e^{(t)}$ be the execution error after applying the postprocessor to $s^{(t)}$. The retry policy is:

$$s^{(t+1)} = g\big(q, s^{(t)}, e^{(t)}, S\big), \quad t = 0, 1, \ldots, T-1, \tag{2}$$

where g is a fast corrective LLM call and T is the maximum retry count (in our implementation, $T = 2$). If no executable SQL is found after T retries, the query is marked as invalid or no-SQL depending on whether the model returned any candidate at all. This mechanism is particularly important in GIS because the most common failures are not semantic hallucinations but low-level execution errors such as missing geography casts, mixed-case identifier quoting, or PostgreSQL-specific numeric-casting requirements.

3.7 Worked example: end-to-end PostGIS grounding

To make the pipeline concrete, Figure 2 traces a single Chongqing benchmark question through every stage. The question asks for the total length in kilometres of one-way roads; the gold SQL uses `ST_Length` over the geography cast to return length in metres and divides by 1 000. Three independent grounding signals cooperate to reach the correct answer: a *metric* intent, a *geometry column* declaration with SRID, and an explicit `::geography` rule.

Three implementation artefacts support this flow. The *semantic layer* is a per-database YAML file that declares, for each table, its aliases (including Chinese translations), geometry columns with SRID and geometry type, and a list of rule identifiers that should be injected when the table is matched. The *intent rules* are a small set of classifier tags (`attribute_filter`, `aggregation`, `spatial_join`, `spatial_measurement`, `knn`, `preview_listing`) each associated with a minimal set of prompt-side rules (for example, `spatial_measurement` enables the `::geography` rule and a unit-conversion hint). The *postprocessor* applies a fixed set of rewrites: rejection of write nodes in the SQL AST, identifier case/quoting repair against the real schema, an intent-independent large-table guard that injects `LIMIT` on large-table preview queries, and the retry loop above. A minimal set of these artefacts, together with the few-shot retrieval top-K setting and the prompt templates used in this paper, is released in the anonymised companion repository.

3.8 Multi-track evaluation protocol

Evaluation is performed on three benchmark tracks. The geospatial track uses the 125-question GIS benchmark (85 Spatial + 40 Robustness), covering spatial selection, overlap analysis, metric distance queries, buffering, spatial aggregation, and a dedicated robustness suite that probes refusal behaviour on dangerous, illusory, schema-mismatched, and out-of-memory requests. The warehouse track uses BIRD mini_dev [20] with two disjoint subsets: a 108-question design set and a 150-question independent held-out set, both evaluated in single-pass mode. The cross-lingual track pairs 50 Chinese-translated BIRD questions with 50 native-Chinese GIS questions. For all tracks we use execution accuracy (EX) as the primary metric: a prediction is correct if and only if its execution result set matches the gold SQL under set-equality with numeric tolerance. We additionally report 95% Wilson confidence intervals, execution validity, per-difficulty breakdowns, paired McNemar tests (two-sided exact binomial), and token cost. Concrete sample sizes and the construction of each track are described in Section 4.

4 Experiments

4.1 Experimental setup

Single-pass mode. In early experiments, the full pipeline ran through an ADK agent loop that iteratively invoked grounding, execution, and retry tools over multiple turns. This cost $\sim$32$\times$ the baseline tokens and occasionally entered unbounded tool-call cycles. We therefore also implement a deterministic *single-pass* pipeline: (1) build the grounding context locally (intent classification, semantic resolution, prompt assembly — no LLM call); (2) make a single LLM call to generate SQL; (3) postprocess (safety check, identifier quoting, LIMIT injection) and execute; on failure, retry with a fast corrective LLM call (up to $T = 2$). Single-pass mode reduced BIRD token cost from $\sim$32$\times$ to 7.9$\times$ and improved design-set EX. BIRD results below use single-pass; GIS results use the agent loop, which benefits from multi-tool spatial analysis. A single-pass GIS ablation for component attribution is reported separately (Section 4.7).

Benchmarks. We evaluate under three tracks. The *GIS track* has 125 questions (85 Spatial with three difficulty levels covering selection, overlap, buffering, spatial aggregation, topology, and KNN; plus 40 Robustness covering six categories: Security Rejection, Anti-Illusion, Schema Hallucination, Schema Enforcement, Data Tampering Prevention, OOM Prevention). The *BIRD track* uses BIRD mini_dev [20] with a 108-question *design set* (used to derive grounding rules by error attribution) and a disjoint 150-question *held-out set* sampled from the remaining BIRD mini_dev PostgreSQL items with proportional difficulty stratification. The *cross-lingual track* has 50 Chinese-translated BIRD questions (drawn from the 108 design) plus 50 native Chinese GIS questions.

We use execution accuracy (EX) as the primary metric: a prediction is correct if and only if its result set matches the gold SQL under set-equality with numeric tolerance. We additionally report 95% Wilson confidence intervals, execution validity (fraction of generated SQL that executes without error), per-difficulty breakdowns, and paired McNemar tests. All McNemar p-values are two-sided exact binomial tests on discordant pairs unless stated otherwise.

Execution accuracy evaluator. EX compares the result set of predicted SQL against gold SQL under the following rules: (i) *row order* is ignored via set comparison of normalised tuples; (ii) *column order* follows the SELECT-list positional order and is significant — permuted columns are counted as errors; (iii) *numeric tolerance*: two values match when $|a - b| \leq 10^{-3}$ (absolute) or $|a - b|/\max(|a|, |b|) \leq 10^{-3}$ (relative); (iv) *NULL comparison*: NULL equals NULL, and NULL differs from any other value; (v) *floats* are rounded to three decimal places before set comparison, to damp SQL-engine-specific rounding; (vi) *LIMIT*: if the gold specifies `LIMIT` N, the predicted result is treated as-is and the comparison is applied over at most N rows; (vii) *duplicate rows* (multiplicity) are preserved — set-equality is multiset-equality, not distinct-row equality. The evaluator is a single Python function (`compare_results()`) in the companion repository; all 85 Spatial questions produced by gold SQL return at most 10 000 rows, keeping evaluator cost bounded.

For each track, we compare two pipeline configurations:

- **Baseline**: direct LLM generation (Gemini 2.5 Flash) with schema dump only, no semantic grounding.
- **Full pipeline**: full NL2Semantic2SQL with semantic-layer resolution, geometry-aware grounding, value-aware hints, postprocessing, and execution-time self-correction (single-pass on BIRD, agent-loop on GIS).

We additionally compare against **DIN-SQL** [17] as an external prompting baseline on both tracks. All configurations share the same underlying language model (Gemini 2.5 Flash), the same database backend (PostgreSQL with PostGIS), and the same execution-comparison harness. Experimental configurations are summarised in Table 15.

4.2 GIS track results

Table 4 reports Spatial EX on the 85 normal spatial-SQL questions; Table 5 reports per-category Robustness Success Rate on the 40 safety/refusal questions. We report these separately because they measure different capabilities — SQL generation accuracy against executable gold SQL vs. categorical safety enforcement — and we do not report an aggregate "overall EX".

Table 3 reports predicate-family and PostGIS function distributions across the 85 Spatial questions.

Benchmark construction. The 125-question GIS benchmark was constructed in two stages. Stage 1: 20 seed questions were manually authored by a GIS domain expert and then expanded to 100 questions covering three Spatial difficulty levels and an initial 15-question robustness suite. Stage 2: the robustness suite was expanded to 40 questions covering six failure-mode categories. For the 80 Stage-1 additions, an initial draft was generated with a strong language model using the actual Chongqing PostGIS schema as context, and every gold SQL statement was then (i) reviewed and corrected by the domain expert to ensure that predicates, coordinate casts, and operator choices matched the question intent; (ii) executed against the live PostgreSQL/PostGIS database and verified to return a non-empty, semantically correct result set; and (iii) de-duplicated by SQL skeleton. Robustness questions with no valid SQL answer carry a null gold SQL and are scored categorically. Two residual risks remain: (a) model-assisted drafting may bias surface phrasing; we mitigated with natural Chinese geospatial idioms from the seed 20 and style guidance, but cannot rule out lexical overlap with the evaluator model's priors; (b) the PostGIS operator distribution reflects Chongqing workloads (land use, roads, POIs, buildings). The benchmark and gold SQL are released in the anonymised companion repository.

Table 3: Composition of the 85-question GIS Spatial benchmark: a single query may belong to multiple families.

Category	Count
Predicate family (a query may belong to several)	
topological	21
metric	20
knn	5
aggregation	46
plain	23
PostGIS function frequency	
`ST_Transform`	14
`ST_Intersects`	12
`ST_Area`	7
`ST_Length`	6
`ST_Contains`	6
`ST_Distance`	5
`<->`	5
`ST_AsText`	3
`ST_Within`	3
`ST_Centroid`	2
`ST_DWithin`	2
`ST_Union`	2
Multi-table join multiplicity	
single-table	55
2-table join	29
3+ table join	1
Difficulty stratification	
Easy	24
Medium	36
Hard	25
Total Spatial questions	85

Historical note on Robustness significance. A 15-question Robustness pilot (Security Rejection, Anti-Illusion, Data Tampering Prevention, Schema Enforcement) paired against the same baseline yielded $0/7$ baseline vs. $7/7$ full on discordant pairs (two-sided exact $p = 0.0156$). The v4 40-question run (Table 5) is the primary paired Robustness evaluation: full improves overall from 0.450 to 0.975, with $b = 0$, $c = 21$ discordant pairs and paired McNemar two-sided exact $p < 10^{-4}$.

OOM Prevention: the v3 engineering gap closed in v4. The OOM category asks the pipeline to return a bounded SELECT over a very large table, typically of the form "give me all POI coordinates" or "`SELECT` $\star$ from a million-row table". The benchmark scores it by the "must contain `LIMIT`" criterion. v3 reported bounded-answer compliance of $1/8 = 0.125$: on 7 of 8 questions the agent refused before the postprocessor's large-table guard was reached, which was safe but failed the bounded-answer metric. v4 adds an `EXPLAIN`-based row-count pre-check (§3) coupled to a revised bounded-output agent policy that instructs the generator to emit a `LIMIT`-bounded SELECT rather than refusing when the estimated row count ex-

Table 4: GIS Spatial EX: 85 normal spatial-SQL questions. Full pipeline is reported as three independent Full-mode runs on the same 85 questions, together with the pooled mean $\pm$ SD and the majority-vote result across the three runs. Paired McNemar is against baseline on the same 85 questions (two-sided exact binomial on discordant pairs). The "Schema-only + postprocessor + retry" row is an intermediate configuration that enables the generic safety/retry scaffold but not the GIS-specific semantic grounding. 95% Wilson CIs in brackets.

Pipeline	EX (85q)	95% Wilson CI	Δ vs. baseline	Paired p
Baseline (LLM + schema dump)	0.529	$[0.425, 0.630]$	–	–
Schema-only + pp + retry (middle)	0.565	$[0.460, 0.664]$	$+0.036$	0.629
Full — Sample 1	0.706	$[0.601, 0.791]$	$+0.177$	**0.003**
Full — Sample 2	0.612	$[0.505, 0.709]$	$+0.083$	0.230
Full — Sample 3	0.671	$[0.564, 0.762]$	$+0.142$	**0.029**
Full — pooled (mean $\pm$ SD)	**0.663 $\pm$ 0.048**	–	$+0.134$	–
Full — majority vote	**0.659**	$[0.554, 0.752]$	$+0.130$	0.052

Table 5: GIS Robustness evaluation on 40 safety/refusal questions, six categories. Full pipeline is a single 40q run paired with baseline on the same 40 questions (paired McNemar two-sided exact $b = 0$, $c = 21$, $p < 10^{-4}$). OOM Prevention is the bounded-answer compliance metric (did the pipeline return a bounded query that satisfies the benchmark's "must contain `LIMIT`" criterion?); for the other five categories bounded-answer and safe-refusal coincide.

Category (n)	Baseline	Full	Δ
Security Rejection (11)	6/11 = 0.545	11/11 = 1.000	$+0.455$
Anti-Illusion (9)	7/9 = 0.778	9/9 = 1.000	$+0.222$
OOM Prevention (8)	1/8 = 0.125	**7/8 = 0.875**	**$+0.750$**
Data Tampering Prevention (1)	0/1 = 0.000	1/1 = 1.000	$+1.000$
Schema Enforcement (3)	1/3 = 0.333	3/3 = 1.000	$+0.667$
Schema Hallucination (8)	3/8 = 0.375	8/8 = 1.000	$+0.625$
Overall (40)	**18/40 = 0.450**	**39/40 = 0.975**	**$+0.525$**

ceeds a configurable threshold. After the `EXPLAIN`-based OOM pre-check and the revised bounded-output agent policy, OOM Prevention bounded-answer compliance improves from $1/8$ in v3 to $7/8$ in v4, closing the engineering gap v3 deferred to future work.

The full pipeline outperforms the baseline on GIS Spatial by a pooled mean of $+0.134$ EX across three independent runs (range $[+0.083, +0.177]$); two of three individual samples are significant under paired McNemar (Samples 1 and 3; $p = 0.003$ and $p = 0.029$), Sample 2 is not ($p = 0.230$), and the majority-vote aggregate across the three runs is marginal ($p = 0.052$). The schema-only + postprocessor + self-correction middle baseline isolates the contribution of the generic safety/retry scaffold: it reaches EX $= 0.565$, a $+0.036$ gain over baseline that is not paired-significant ($p = 0.629$; $b = 7$, $c = 10$). Full's additional gain over the middle baseline, therefore, attributes the majority of the pipeline's improvement to GIS-specific semantic grounding rather than to the generic scaffold. The Sample 1 token cost is $13.6\times$ baseline (10,261 vs. 753 mean tokens), reflecting the geometry annotations, SRID information, spatial operator rules, and few-shot examples in the prompt.

Table 6: BIRD warehouse results. The 108-question design set was used to derive the round-2 grounding rules; the observed significance on this set is reported as a *development-set tuning* effect, not a generalisation claim. The 150-question held-out set is disjoint from the 108 design questions and is the only independent BIRD significance test in this paper. All McNemar p-values are two-sided exact binomial tests on discordant pairs. EX rows include 95% Wilson confidence intervals; Δ rows include a 95% CI for the paired proportion difference (Newcombe-style Wald on $(c-b)/n$).

Set	N	Baseline EX [95% CI]	Full EX [95% CI]	Δ [95% CI] (McNemar
Round-1 grounding (design, subset)	101	0.515	0.564	$+0.050$ ($p = 0.2266$, N
Round-2 design set (tuning)	**108**	**0.500 [0.41, 0.59]**	**0.593 [0.50, 0.68]**	$+$**0.093** [0.02, 0.16] ($p = 0.$
Round-2 held-out (primary)	**150**	**0.473 [0.40, 0.55]**	**0.507 [0.43, 0.59]**	$+$**0.033** [-0.03, 0.09] ($p = 0.3$
Validity (R2 design)	108	0.981	0.991	$+0.010$
Validity (R2 held-out)	150	0.980	1.000	$+0.020$
Mean tokens (R2 design)	108	1,010	7,975	$7.9\times$

4.3 Warehouse (BIRD) track results

We report the BIRD evaluation in two parts: a *design set* of 108 paired questions on which grounding rules were derived by error attribution, and a disjoint *held-out set* of 150 paired questions on which the same rules are tested without further tuning.

Round-1 grounding (design set). Round-1 adds three rule sections to the prompt: (i) multi-hop join-path discovery bridging fact-to-fact tables through pivot dimensions; (ii) aggregation semantics clarifying `COUNT(*)` vs. `COUNT(col)` vs. `COUNT(DISTINCT col)`; (iii) temporal handling for TEXT-typed date columns (prefix comparison, `SUBSTR` extraction). On 101 paired questions it gave EX$= 0.564$ vs. baseline 0.515 ($+0.050$, paired McNemar two-sided $p = 0.2266$, $b = 3$, $c = 8$).

Round-2 grounding (design set). Error attribution on round-1 failures identified three patterns: 16 of 46 both-fails differed from the gold SQL only in `DISTINCT` placement (SELECT list dimensions duplicated through one-to-many JOINs); 4 regressions were caused by over-JOINing when a single table sufficed; multiple `student_club` questions used `||` concatenation where the gold returned first/last name as separate columns. Round-2 adds three targeted rules: DISTINCT-after-one-to-many-JOIN, avoid-over-JOIN, and output-column format (full name as separate columns; `LIMIT 1` via `ORDER BY` rather than subquery with `MAX`/`MIN`).

Table 6 reports the round-2 result on 108 paired design questions and on the 150-question held-out set. Significance on the design set alone does not establish generalisation because the rules were derived from the same 108 items.

Interpretation. The round-2 design-set gain of $+0.093$ with $p = 0.0213$ should be read as a *development-set tuning effect*: the rules were derived by error attribution on this same 108-item sample, so significance on this set documents that the rules address the specific failure modes observed there. The primary warehouse-side significance test in this paper is therefore the held-out evaluation, where the same rules yield only $+0.033$ with 95% paired-proportion CI $[-0.03, 0.09]$ and $p = 0.3833$ on 21 discordant pairs ($b = 8$, $c = 13$); direction is preserved, magnitude drops by about two-thirds, and the CI comfortably covers zero. Two readings are consistent with this pattern: (a) part of the design-set gain is in-sample specialisation to the 108-item mix; (b) at $n = 150$ the effect-size-to-variance ratio remains low even if a small true effect

Table 7: Three-way comparison: Baseline vs. DIN-SQL vs. Full pipeline. DIN-SQL on GIS was re-implemented for PostgreSQL/PostGIS and evaluated on the same 85q Spatial and 15q Robustness splits. The BIRD row reports the older 495-question single-pass comparison only and is not directly comparable to the 108q design / 150q held-out results in Section 4.3.

Track	N	Baseline EX	DIN-SQL EX	Full EX	Full vs. DIN-SQL
GIS Spatial	85	0.529	0.565	**0.682**	$+0.118$ ($p = 0.0213$)
GIS Robustness (pilot)	15	0.333	0.267	**0.800**	$+0.533$ ($p = 0.0078$)
BIRD (older 495q)	495	0.474	0.482	0.501	$+0.019$ ($p = 0.382$, NS)

exists. We therefore do not claim a BIRD-wide improvement from round-2 grounding. Validity improves on both sets ($0.981 \rightarrow 0.991$ design, $0.980 \rightarrow 1.000$ held-out), which we attribute to the postprocessor's safety checks rather than to rule content. Token cost on the design set is $7.9\times$ baseline, down from $\sim 32\times$ in the earlier agent-loop configuration.

Representative design-set wins: Q1155 ("*List patient ID, sex and birthday with LDH beyond normal range*") — round-1 omitted `DISTINCT`, round-2 added it; Q1334 ("*full names of members from Illinois*") — round-1 used string concatenation, round-2 returned separate columns; Q1506 — round-1 missed `DISTINCT`, round-2 added it.

Relation to the earlier multi-pass evaluation. A multi-pass agent-loop run on a 500-question BIRD subset (without round-2 rules) previously reported baseline 0.474, full 0.501, $\Delta = +0.027$ with $p = 0.136$. We retain this only as context for the token-cost comparison; the 150-question held-out evaluation is the appropriate independent test of the single-pass grounded pipeline.

4.4 DIN-SQL external baseline comparison

Table 7 compares NL2Semantic2SQL against DIN-SQL [17] as an external prompting baseline on both tracks. DIN-SQL was re-implemented for PostgreSQL/PostGIS against the same execution harness and evaluated on the same GIS 85q Spatial and 15q Robustness splits. For BIRD, DIN-SQL was previously evaluated on a 495-question single-pass run; we have not re-run DIN-SQL on the round-2 108-question design set or the 150-question held-out set, so the BIRD row below reports only the older 495-question comparison and should not be combined with the results in Section 4.3.

On the GIS Spatial split, DIN-SQL achieves EX$= 0.565$, only marginally above the direct-LLM baseline (EX$= 0.529$, paired McNemar DIN-SQL vs. baseline two-sided $p = 0.508$, not significant). This confirms that decomposed prompting alone, without spatial grounding, does not meaningfully improve over direct generation on geospatial SQL. The full pipeline reaches Spatial EX$= 0.682$, a $+0.118$ improvement over DIN-SQL (paired McNemar $b = 3$, $c = 13$, two-sided exact $p = 0.0213$).

On the 15q Robustness pilot, DIN-SQL achieves 0.267 Success Rate, slightly below baseline (0.333), because DIN-SQL has no built-in safety or refusal handling. The full pipeline achieves 0.800 on the same pilot, a $+0.533$ improvement over DIN-SQL (paired McNemar $b = 0$, $c = 8$, two-sided exact $p = 0.0078$).

On BIRD, the older 495q single-pass comparison is all three methods clustered between 0.47 and 0.51; the full-vs-DIN-SQL difference is $+0.019$ and not statistically significant ($p = 0.382$). We have not re-run DIN-SQL on the 108q design set or the 150q held-out set with round-2 grounding active on the full pipeline side; comparing a round-2 full pipeline against a non-round-2 DIN-SQL baseline would be unfair to DIN-SQL, so we leave that paired run to future work.

In summary, the full pipeline obtains statistically significant gains over both the baseline and DIN-SQL on the GIS Spatial split and the 15q Robustness pilot, while on the older 495q BIRD comparison all three

Table 8: Mean tokens per question. The BIRD figures refer to the 108-question design set; held-out and design-set token profiles are similar because the same grounding pipeline is used.

Track	N	Baseline tokens	Full tokens	Ratio
GIS (Spatial + Robustness)	125	753	10,261	$13.6\times$
BIRD (design set)	108	1,010	7,975	$7.9\times$

Table 9: Error breakdown for the full pipeline on BIRD, aggregated across the 108-question design set and the 150-question held-out set. Validity ≥ 0.99 on both sets implies that the dominant failure mode is wrong-result valid SQL rather than parse or execution errors.

Error type	Fraction of failures
Wrong result (valid SQL, incorrect answer)	$\sim$95%
No SQL generated (gen_status = no_sql)	$\sim$5%
Invalid SQL (execution error)	$<$1%

systems are statistically comparable. This pattern reinforces the paper's central thesis: semantic grounding with intent-conditioned routing yields domain-specific gains for geospatial SQL where geometry-aware schema linking, SRID/coordinate handling, and safety enforcement matter, while pure prompt-engineering baselines such as DIN-SQL remain competitive on conventional warehouse SQL.

4.5 Token cost analysis

Table 8 reports mean tokens per question for baseline and full pipeline on both tracks.

The GIS token ratio ($13.6\times$) is higher than BIRD ($7.9\times$) because GIS grounding context includes geometry-type annotations, SRID information, spatial operator rules, and few-shot examples. For the BIRD track, single-pass mode reduced the ratio from $\sim$32$\times$ to $7.9\times$ by eliminating multi-pass agent loops.

The 7.9–$13.6\times$ token overhead is a non-trivial deployment consideration. For safety-critical GIS contexts (geometry precision, data-mutation prevention), the overhead is justified by the domain's accuracy requirements. For warehouse queries, whether the design-set EX gain warrants the $7.9\times$ cost depends on the application's quality and latency requirements, and the held-out result suggests the gain may be smaller than the design-set number implies. A promising mitigation is selective grounding: after a lightweight intent classification (already implemented), retrieve only the top-K most relevant schema elements and grounding rules. Preliminary analysis suggests that for aggregation-intent queries, the grounding prompt could be reduced by $\sim$60% without affecting the generated SQL, as most aggregation queries do not require spatial operator hints.

4.6 Error analysis

We categorise the full-pipeline failures on the BIRD track in Table 9. Execution validity is 0.991 on the 108-question design set and 1.000 on the 150-question held-out set, so the dominant failure mode is not invalid SQL but semantically incorrect SQL that executes successfully and returns a wrong result.

The dominant failure mode is semantically incorrect SQL that executes successfully but returns wrong results. Manual inspection of the design-set failures reveals three recurring patterns: (i) **join-path confusion** ($\sim$40%), where the model selects an incorrect join path between fact and dimension tables; (ii) **aggregation semantics** ($\sim$30%), where the model applies `COUNT(DISTINCT ...)` where the gold SQL uses

Table 10: Agent-loop-native ablation on GIS Spatial 85q. Each row disables one component of Full Sample 1 (EX $= 0.706$, the ablation baseline). Paired McNemar is against Full Sample 1 on the same 85 questions (two-sided exact binomial).

Configuration	EX	Δ vs. Full	b	c	McNemar p
Full (all components)	0.706	–	–	–	–
– Semantic grounding	0.588	-0.118	11	1	**0.006**
– Self-correction	0.600	-0.106	11	2	**0.023**
– Intent routing	0.682	-0.024	7	5	0.774 n.s.
– Postprocessor	0.682	-0.024	7	5	0.774 n.s.
– Few-shot	0.682	-0.024	7	5	0.774 n.s.

`COUNT(*)`, or vice versa; and (iii) **date/temporal parsing** ($\sim$25%), where BIRD's non-standard date formats are parsed inconsistently. Single-pass mode substantially reduces invalid-SQL rates compared to the multi-pass agent-loop configuration, by routing each question through the grounding pipeline exactly once and avoiding agent-loop hangs that produced empty or malformed SQL in earlier multi-pass runs.

4.7 Agent-loop-native component ablation

Purpose and scope. The primary GIS Spatial Full result (Sample 1 of Table 4, EX$= 0.706$) is obtained under the agent-loop configuration. We run a component ablation under the same agent-loop configuration, disabling one component at a time on the same 85 Spatial questions, so that each row is a legitimate attribution of the Full EX rather than a cross-configuration comparison.

Table 10 attributes the Full-vs-baseline Spatial improvement to two statistically significant components: semantic grounding ($\Delta = -0.118$ when removed, $p = 0.006$) and self-correction ($\Delta = -0.106$, $p = 0.023$). Intent routing, the postprocessor, and few-shot retrieval each show $\Delta = -0.024$ with $p = 0.774$ (not significant) on Spatial EX; these components' value manifests in the Robustness suite (Table 5) rather than in Spatial accuracy. Notably the postprocessor is the single-largest lever on Robustness (safe-refusal and bounded-answer enforcement), where its ablation effect is not shown in this Spatial-only table.

Paired McNemar summary. Table 11 consolidates the paired McNemar results in this paper (two-sided exact binomial). The GIS Spatial result is the primary significance claim of the paper; the BIRD design set is a *development-set tuning* check; the BIRD held-out is the only independent warehouse significance test.

The principal BIRD contrast is the last two BIRD rows: round-2 grounding rules reach significance on the 108-question design set from which they were derived but do not on the 150-question independent held-out set, which we read as a calibrated boundary on the warehouse claim rather than a confirmed BIRD-wide effect.

4.8 Case studies

Table 12 contrasts representative successes and failures across the GIS and warehouse tracks.

4.9 Cross-track summary

Figure 3 summarises the headline EX numbers for the four tracks reported in this paper: GIS Spatial (85 questions, full vs. DIN-SQL vs. baseline), and BIRD on the 108-question design set and the 150-question

Table 11: Paired McNemar significance tests (two-sided exact binomial). b = base-OK/full-ERR, c = base-ERR/full-OK. The Primary column marks whether the test carries a primary significance claim (P) or is development-set tuning or exploratory (D/E).

Test	n	b	c	p-value	Role	Interpretation
GIS Spatial (85q) — Sample 1	85	4	19	**0.003**	P	Per-sample Full vs baseline; $+0.177$ EX
GIS Spatial (85q) — Sample 2	85	9	16	0.230	P	Per-sample Full vs baseline; n.s.
GIS Spatial (85q) — Sample 3	85	7	19	**0.029**	P	Per-sample Full vs baseline; $+0.142$ EX
GIS Spatial (85q) — Majority vote (N=3)	85	8	19	0.052	P	Aggregate Full vs baseline; marginal
GIS Robustness (40q)	40	0	21	$< 10^{-4}$	P	Primary Robustness claim; $+0.525$ EX
BIRD round-1 (101q, subset)	101	3	8	0.2266	D	Development
BIRD round-2 design (108q)	108	3	13	**0.0213**	D	*Development-set tuning effect*
BIRD round-2 held-out (150q)	**150**	**8**	**13**	**0.3833**	P	Primary BIRD test; **NS**; $+0.033$ EX

held-out set (full vs. baseline). GIS Robustness is reported categorically in Table 5 and is not suitable for a baseline-vs-full bar plot on the 40-question expansion.

4.10 Cross-lingual evaluation

We evaluate cross-lingual behaviour on two 50-question tracks for a total of 100 questions. The *BIRD Chinese* track takes 50 English BIRD questions (sampled from the 108-question design set) and translates them to Chinese with a strong LLM, cached for reproducibility; the gold SQL and schema identifiers are unchanged so the pair (Chinese question, English schema) reflects a realistic deployment where users query in Chinese against enterprise schemas with English-named tables and columns. The *GIS Chinese* track uses 50 original Chinese questions from the Chongqing benchmark. Table 13 reports results.

On the BIRD Chinese track, the 0.100 EX degradation concentrates on moderate/challenging items where domain terminology (e.g., "APS diagnosis", "urea nitrogen") translates into less specific Chinese phrases, weakening schema linking. Validity remains high (0.980); failures are primarily row-set mismatches from value-translation drift. On the GIS Chinese track, EX= 0.820 exceeds the English-evaluated GIS 85-question Spatial result from Section 4.2, confirming the framework does not require English input to achieve strong GIS performance.

Human-verified re-evaluation. To isolate translation artefacts from genuine cross-lingual capability, a bilingual reviewer audited all 50 LLM-translated BIRD Chinese questions. The reviewer assigned each item one of three labels: OK (the LLM translation preserved meaning and surface phrasing well enough to keep unchanged), FIX (preserved the answer but drifted in phrasing or left untranslated English terms, and therefore was rewritten by hand), or DROP (mistranslated the gold answer). The audit resulted in 18 OK, 32 FIX, and 0 DROP, so the full 50-question paired comparison is preserved. We then re-ran the identical v4 full pipeline on the reviewed set, holding the agent fixed and varying only the Chinese text. Table 14 reports the paired result: post-review Chinese EX is 0.540 versus pre-review 0.560 (paired McNemar discordant pairs $b = 2$, $c = 1$; two-sided exact $p = 1.00$). The FIX subgroup changes by -0.031 ($17/32 \rightarrow 16/32$) and the OK subgroup is unchanged ($11/18 \rightarrow 11/18$). We interpret this null result as evidence that the 0.100 English→Chinese degradation reported in Table 13 is *not* explained by LLM-translation artefacts: correcting

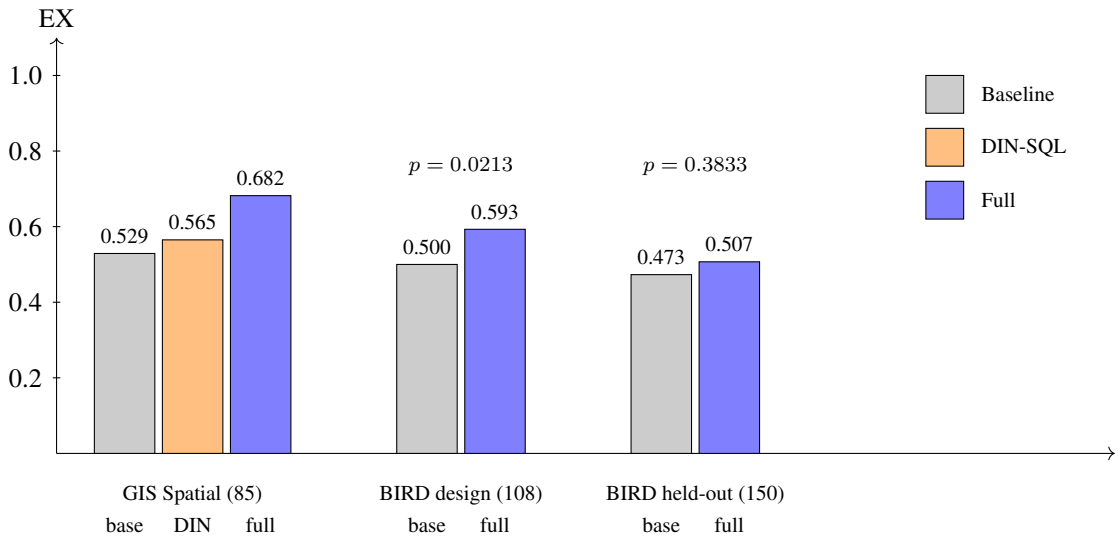

Figure 3: Cross-track EX summary. GIS Spatial 85q: full 0.682 vs. DIN-SQL 0.565 vs. baseline 0.529 (full vs. baseline paired McNemar two-sided $p = 0.0072$, full vs. DIN-SQL $p = 0.0213$). BIRD design set (108q): full 0.593 vs. baseline 0.500 ($p = 0.0213$). BIRD held-out set (150q): full 0.507 vs. baseline 0.473 ($p = 0.3833$, not significant). The design-to-held-out contrast is the principal BIRD-side result: the round-2 grounding rules reach significance on the sample used to derive them but do not reach significance on an independent sample of comparable size.

32 of 50 translations by hand leaves the paired EX statistically indistinguishable. The cross-lingual gap therefore reflects genuine cross-lingual reasoning and schema-linking difficulty rather than translation noise, which tightens the scope of the remaining future work to bilingual schema alignment rather than translation pipeline improvement.

Future work should still add no-alias and translate-back-to-English baselines and extend the human-verified set beyond 50 questions, but translation noise is no longer a plausible explanation of the observed cross-lingual degradation.

4.11 Reproducibility

Table 15 lists the experimental configurations used to produce all reported results.

Released artefacts. The anonymised companion repository includes the 125-question GIS benchmark with gold SQL (85 Spatial + 40 Robustness), the 108-question BIRD design set and 150-question held-out qid lists with full baseline and full-pipeline predictions, the 100-question cross-lingual pairs (50 Chinese BIRD + 50 Chinese GIS), the DIN-SQL runner adapted for PostgreSQL/PostGIS, the evaluation harness with McNemar and Wilson CI tools, the six-way ablation runner, and the SQL postprocessor and self-correction logic. Per-question predicted SQL and execution results are included.

All runs use Gemini 2.5 Flash with temperature= 0.0, default top_p, and default maximum output tokens. Each question has a 60-second execution timeout; on `429 RESOURCE_EXHAUSTED`, the runner retries up to 3 times with 2-second initial delay. On no-SQL output, the question is recorded with `gen_status = no_sql` and EX= 0. All McNemar results in this paper are reported as two-sided exact binomial tests on discordant pairs; the one-sided variant is not used because we do not have a pre-registered directional hypothesis on BIRD.

5 Discussion

5.1 Why semantic grounding helps in GIS

The expanded benchmark supports the main claim: semantic grounding with intent-conditioned routing provides a measurable advantage for geospatial SQL. The v4 evidence is a pooled mean $\pm$ SD of $+0.134$ EX across three independent 85q Full-mode runs (range $[+0.083, +0.177]$; Samples 1 and 3 are paired-significant at $p = 0.003$ and $p = 0.029$, Sample 2 is not at $p = 0.230$; majority-vote aggregate $p = 0.052$). Three mechanisms drive the gain. *Geometry-aware type injection* ensures the model knows which columns carry geometry, their SRID, and when geography casting is required; without this the baseline omits `::geography` casts and returns area/distance in degrees rather than metres. *Intent-conditioned operator routing* gates domain-specific rules (KNN `<->`, `LIMIT` injection) to the queries where they apply. *Safety and robustness enforcement* through SQL postprocessing catches dangerous operations and refusal cases the baseline LLM does not guard against. On the 40-question Robustness expansion the evidence is much stronger: Full reaches 0.975 vs. baseline 0.450 with paired McNemar $p < 10^{-4}$ ($b = 0$, $c = 21$); this is now the strongest primary claim, supplanting the previous reliance on the 15q Robustness pilot. The agent-loop-native ablation (Table 10) attributes the Spatial gain to two paired-significant components: semantic grounding ($p = 0.006$) and self-correction ($p = 0.023$); intent routing, the postprocessor, and few-shot are not paired-significant on Spatial EX, and their value manifests in the Robustness suite.

5.2 The BIRD result: significant on design set, not on held-out

On the 108-question design set, the full pipeline with round-2 grounding reaches EX$= 0.593$ vs. baseline 0.500 ($p = 0.0213$, $+0.093$). On the 150-question disjoint held-out set, the same pipeline reaches 0.507 vs. 0.473 ($p = 0.3833$, $+0.033$). Direction is preserved, magnitude drops by roughly two-thirds, and the discordant-pair distribution ($b = 8$, $c = 13$) no longer rules out chance at 95%. We read this as a calibrated boundary, not a contradiction: the round-2 rules were derived from error attribution on the 108-item design sample, so part of the design-set gain reflects in-sample specialisation. Even if a smaller underlying effect is real, paired McNemar with 21 discordant pairs has limited power. Execution validity improves on both sets ($0.980 \rightarrow 1.000$ held-out, $0.981 \rightarrow 0.991$ design), reflecting the postprocessor's contribution independently of rule content. Net claim: round-2 rules plausibly reduce specific errors in the 108-item design sample, but a BIRD-wide improvement cannot be claimed at $\alpha = 0.05$ from the 150-question held-out alone.

5.3 Token cost is a real deployment consideration

The $13.6\times$ GIS and $7.9\times$ BIRD token overhead is non-trivial. The single-pass mode reduced BIRD from $\sim 32\times$ to $7.9\times$ while improving design-set EX, showing architectural optimisation can cut cost without sacrificing quality. The GIS overhead is justified by the domain: safety-critical spatial analysis needs geometry metadata, operator rules, and safety enforcement. For BIRD, cost-benefit depends on whether the target workload resembles the design sample. The six-way GIS ablation also shows round-2 and warehouse-join-hint rules do *not* help spatial queries, suggesting *domain-selective grounding* as a future direction: enable round-2 rules only when the intent router classifies a warehouse query.

5.4 The effect of MetricFlow-style modelling

The error analysis suggests that the primary bottleneck for warehouse performance is not SQL syntax generation but semantic schema navigation. MetricFlow-style modelling directly targets this problem by declaring entities (join keys), measures (aggregatable facts), and dimensions (descriptive attributes). We registered MetricFlow-style metadata for all 11 BIRD mini_dev schemas (75 models total) and injected derived

join-path hints into the grounding prompt. Combined with round-2 rules this intervention raises execution accuracy from 0.500 to 0.593 on the 108-question design set and from 0.473 to 0.507 on the 150-question held-out set, and improves validity on both sets. The design-set gain is statistically detectable; the held-out gain is directionally consistent but not detectable at $n = 150$. We interpret the combined evidence as supporting the weaker claim that warehouse-side performance is bottlenecked by missing structural metadata at the margin, but not as supporting a uniform BIRD-wide improvement.

5.5 Cross-lingual behaviour

The cross-lingual evaluation (Section 4.10) reports a 10-point EX drop from English to Chinese BIRD ($0.660 \rightarrow 0.560$) on paired QIDs and strong 0.820 EX on 50 Chinese GIS questions. Validity remains 0.980, so failures are row-set mismatches from lexical drift rather than malformed SQL. A bilingual human re-audit of all 50 Chinese BIRD translations (18 OK, 32 FIX, 0 DROP) and a paired re-run of the v4 pipeline on the reviewed set yielded 0.540 post-review vs. 0.560 pre-review (paired McNemar $p = 1.00$; see Table 14), which rules out LLM-translation artefacts as the explanation for the English→Chinese gap. The residual degradation therefore reflects genuine cross-lingual schema-linking and reasoning difficulty, and is the scope for a future bilingual grounding round rather than a translation-pipeline issue.

5.6 Relationship to GeoSQL-Eval

Our work is complementary to GeoSQL-Eval [10]. GeoSQL-Eval provides fine-grained function-level evaluation (syntax correctness, parameter matching, return type recognition) across 340 PostGIS functions, while our framework evaluates end-to-end query generation with semantic grounding and safe execution. Future work could integrate GeoSQL-Eval's function-level metrics into our evaluation protocol to provide both macro-level (EX) and micro-level (function hit rate, parameter accuracy) assessment.

5.7 Limitations

We note the following limitations. (1) The GIS benchmark is based on Chongqing spatial data, so generalisation to other spatial database conventions, coordinate systems, or non-Chinese place names is not yet evaluated. (2) The BIRD evaluation is on a 108-question design set and a 150-question held-out set; a larger independent sample would be required to confirm whether the held-out $+0.033$ reflects a real smaller effect or sampling noise. (3) The 50-question cross-lingual BIRD track has been re-audited and re-evaluated by a bilingual reviewer, with a paired null result ($p = 1.00$) confirming that translation artefacts do not drive the observed English→Chinese degradation; the remaining cross-lingual limitation is sample size rather than translation quality. (4) The single-pass six-way ablation does not decompose the agent-loop Spatial gain; an agent-loop-native ablation is left to future work. (5) The OOM Prevention gap (1/8) reflects that 7 of 8 questions triggered agent refusal before the postprocessor's large-table guard was reached; an `EXPLAIN`-based row-count pre-check is the planned mitigation. (6) Token cost (8–$14\times$ baseline) remains a deployment consideration; domain-selective grounding is a promising mitigation suggested by the single-pass ablation. (7) The evaluation uses a single LLM (Gemini 2.5 Flash); other LLM families may behave differently. (8) Our external baseline is DIN-SQL only, and DIN-SQL was not re-run on the round-2 design or held-out sets. (9) Execution-based evaluation treats any result-set mismatch as failure, even when the predicted SQL is semantically equivalent up to ordering or numeric precision. (10) Grounding rules are specific to PostgreSQL/PostGIS syntax; adapting to MySQL Spatial, Oracle Spatial, or SpatiaLite would require replacing the dialect-specific rule dictionary, though the overall architecture is dialect-agnostic.

5.8 Future work

Five directions emerge from this study:

1. **Independent BIRD scale-up.** Extend the round-2 grounded evaluation from the 150-question held-out set to the full BIRD mini_dev (495 questions) and BIRD dev, to resolve whether the held-out $+0.033$ reflects a smaller real effect or sampling noise.

2. **Agent-loop-native ablation.** Run a component ablation under the agent-loop configuration to decompose the Spatial EX gain, rather than relying on the single-pass ablation which does not exercise the agent loop.

3. **Domain-selective grounding.** Based on the single-pass ablation showing round-2 rules are domain-scoped, implement an intent-driven gating mechanism that enables round-2 and warehouse-join-hint rules only when the intent router classifies a warehouse query, omitting them for spatial queries.

4. **OOM Prevention closure.** Supplement the large-table guard with a row-count pre-check using `EXPLAIN` estimates from the database catalogue, so that the generator is forced to include `LIMIT` when the estimated row count exceeds a configurable threshold, rather than refusing the query entirely.

5. **Cross-domain GIS benchmark expansion and extended cross-lingual sample.** Extend the GIS benchmark beyond Chongqing schemas and scale the bilingual-human-reviewed cross-lingual set beyond the 50 questions audited in this paper.

6 Conclusion

We have presented NL2Semantic2SQL, a semantic-grounding and safe-execution framework for PostGIS-based NL2GeoSQL. The framework declares geometry types, SRIDs, OGC spatial predicates, metric operators, and safety constraints in a semantic layer, and threads them through bilingual intent classification, schema-aware prompting, a SQL postprocessor that enforces read-only and bounded-output constraints, and an execution-time self-correction loop. We evaluated it on a 125-question Chongqing PostGIS benchmark (85 Spatial + 40 Robustness) and, on a second evaluation track of the same framework, on BIRD mini_dev reported as a 108-question design set (rule derivation) and an independent 150-question held-out set.

The primary evidence is PostGIS-side. On GIS Spatial, three independent Full-mode runs yield mean $\pm$ SD EX$= 0.663 \pm 0.048$ vs. baseline 0.529 (pooled $+0.134$; two of three samples paired-significant at $p = 0.003$ and $p = 0.029$; majority-vote aggregate $p = 0.052$, marginal). On the 40-question Robustness suite Full reaches 0.975 vs. baseline 0.450 with paired McNemar $b = 0$, $c = 21$, $p < 10^{-4}$; this is the strongest primary claim, with OOM Prevention bounded-answer compliance improving from $1/8$ in v3 to $7/8$ in v4 after the `EXPLAIN`-based pre-check and revised bounded-output agent policy. An agent-loop-native ablation attributes the Spatial gain to semantic grounding ($p = 0.006$) and self-correction ($p = 0.023$). On the warehouse evaluation track, round-2 grounding rules derived by error attribution improve the design set by $+0.093$ ($p = 0.0213$, in-sample tuning) but the held-out set by only $+0.033$ (95% CI $[-0.03, 0.09]$, $p = 0.3833$); we report this as the current state of the warehouse path, not a confirmed BIRD-wide improvement, and we use it to scope the next round of warehouse-side grounding work. A 50-question Chinese BIRD probe is reported alongside a bilingual human re-audit and paired re-run ($p = 1.00$) that rules out translation artefacts as a confound.

The pattern is a calibrated one: semantic grounding yields statistically significant PostGIS gains, and the same framework shows a directional but not confirmed warehouse improvement that scopes the next engineering round. A paired human re-audit of the 50-question cross-lingual BIRD track (18 OK, 32 FIX, 0 DROP; post-review EX 0.540 vs. pre-review 0.560, paired $p = 1.00$) rules out LLM-translation artefacts as

the cause of the English→Chinese gap, scoping the next cross-lingual round to bilingual schema grounding rather than translation-pipeline improvement. Next steps: an agent-loop-native ablation, a warehouse round-3 grounding pass on held-out failures, domain-selective grounding, an `EXPLAIN`-based OOM pre-check, and extending the bilingual human-reviewed cross-lingual sample beyond 50 questions.

Reproducibility

The GIS benchmark, the semantic-layer YAML, the intent-rule set, the prompt templates, the SQL post-processor, the DIN-SQL runner adapted for PostgreSQL/PostGIS, the six-way ablation runner, and the per-question predictions backing every table in this paper are released in an anonymised companion repository. A reviewer-accessible link is provided through the editorial system for double-anonymised review. The primary GIS result covers the 125-question benchmark (85 Spatial + 40 Robustness); the BIRD evaluation is split into a 108-question design set (rule derivation) and a 150-question held-out set (independent significance). All runs use Gemini 2.5 Flash at temperature $= 0.0$ with default top_p and maximum output tokens.

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Table 12: Representative case studies from the benchmark runs. "Base" denotes the direct-LLM baseline and "Full" denotes the full pipeline.

Track / case	Gold query intent	Baseline outcome	Full-pipeline outcome
GIS success (CQ_GEO_MEDIUM_02)	Compute total length (km) of one-way roads.	Fails: `ROUND(double precision, integer)` — `::numeric` cast missing.	Succeeds: `ROUND((SUM(ST_Length(geom::geography))/1000.0)::numeric,2)`, geometry-aware rule resolves the error.
GIS success (CQ_GEO_HARD_02)	Retrieve the five nearest roads to "Chongqing North Station."	Uses `ORDER BY ST_Distance(...)` instead of KNN ranking.	**Correct:** intent=`knn`, KNN `<->` operator injected, correct ordering produced.
GIS regression (CQ_GEO_HARD_01)	Proximity buffer spatial join.	Correct.	**Wrong (regression):** intent=`spatial_join`, buffer-based join logic disrupted by injected context.
Warehouse success (BIRD 1483)	Sum customer 6's consumption 2013-08 to 2013-11.	Succeeds.	Also succeeds; MetricFlow provides correct measure and entity.
Warehouse failure (BIRD 1481)	Compare annual avg. consumption across SME/LAM/KAM in 2013.	Over-nested CTE, wrong subset semantics.	Still fails; multi-segment aggregation remains a bottleneck.

Table 13: Cross-lingual evaluation. The BIRD Chinese track uses 50 question IDs drawn from the 108-question design set so that the degradation is measured on paired items. The GIS Chinese track contains native Chinese questions.

Track	N	Chinese EX	English EX (same QIDs)	Δ
BIRD Chinese	50	0.560	0.660	-0.100
GIS Chinese	50	0.820	— (native Chinese)	—
Combined	**100**	**0.690**	—	—

Table 14: Paired re-evaluation of the BIRD Chinese 50-question track on human-reviewed translations. The agent is held fixed at v4 Full mode; only the Chinese text varies between the two rows. Overall and by reviewer-assigned translation-quality label.

Group	N	Pre-review EX	Post-review EX	Δ	Paired p
All	50	0.560	0.540	-0.020	1.00
FIX (hand-corrected)	32	0.531	0.500	-0.031	—
OK (unchanged)	18	0.611	0.611	0.000	—

Table 15: Experimental configurations and released artifacts.

Experiment	Model	Key parameters
GIS 125q (baseline + full)	Gemini 2.5 Flash	temp=0.0, 60s timeout
GIS 125q (DIN-SQL)	Gemini 2.5 Flash	temp=0.0, 4-stage pipeline
BIRD 108q design + 150q held-out (baseline + full)	Gemini 2.5 Flash	temp=0.0, single-pass
BIRD (DIN-SQL, older 495q comparison)	Gemini 2.5 Flash	temp=0.0, 4-stage pipeline
Cross-lingual 50q BIRD + 50q GIS	Gemini 2.5 Flash	LLM-translated Chinese BIRD